ISCED Code: 0714	CCE-3509	Course Title: Computer Architecture		
Credit Hours: 2	**Contact Hours: 2 CH per Week**	**Prerequisite Course:**		
Course Assessments	**CIE: Continuous Internal Evaluation**	**Attendance**		**10 Marks**
		Class Test/ Assignment/ Quizzes		**10 Marks**
		Mid-term		**30 Marks**
	SEE: Semester End Examination			**50 Marks**

Course Objective: This course is designed to introduce students to the basic concepts of computers, their design and how they work. It encompasses the definition of the machine's instruction set architecture, its use in creating a program, and its implementation in hardware. The course addresses the bridge between gate logic and executable software, and includes programming both in assembly language (representing software) and HDL (representing hardware).

Course Learning Outcomes (CLOs):

CLOs	CLOs Upon the successful completion of the course, students will be able to	Bloom's Taxonomy Domain/Level	Program Learning Outcomes (PLOs)
CLO: 1	Understand the Overview, Computer System, Arithmetic and logic, Central processing unit and parallel organization.	Cognitive (Understanding)	PLO – 1
CLO: 2	Understand the Computer and Processor Design, Hazards; Exceptions; external and internal memory Pipeline and multiple processor systems.	Cognitive (Understanding)	PLO – 2
CLO: 3	Develop and design an instruction set architecture and subsystems of central processing unit.	Cognitive (Evaluating)	PLO – 3

Course Contents:

Contents	CLOs	Lecture	Tutorial	Practical
Mid Term Examination (30 Marks)				
Fundamentals of Computer Organization and Architecture: Fundamentals of computer Design, Processor Design, Computer Evolution and Performance, Processor Design;	CLO – 1, 2, & 3	3	0	0
Computer Function and Interconnection: overview of computer BUS standards;		2	0	0
Multiprocessors: types of multiprocessors, performance, single bus multiprocessors, multiprocessors connected by network, clusters;		2	0	0
Cache Memory: Computer Memory System Overview, Cache Memory Principles, Elements of Cache Design, Pentium 4 Cache Organization, ARM Cache Organization;		3	0	0
Internal Memory: Memory organization, ARM Cache Organization, cache, Error Correction, virtual memory, channels; Concepts of DMA and Interrupts, Advanced DRAM Organization;		3	0	0
External Memory: Magnetic Disk, RAID, Solid State Drives, Optical Memory, Magnetic Tape; Input/ Output: External Devices, I/O Modules, Programmed I/O, Interrupt Driven I/O, Direct Memory Access, I/O Channels and Processors, Thunderbolt and Infinite Band;		2	0	0
Semester End Examination (SEE) = 50 Marks **Part – A (20 Marks)**				
Operating System Support: Operating System Overview, Scheduling, Memory Management, Pentium Memory Management, ARM Memory Management; Number Systems, Computer Arithmetic, Machine Instruction Characteristics, Types of Operands, Types of Operations; Processor Structure and Function;	CLO – 1, 2, & 3	5	0	0
Processor design: data paths – single-cycle and multi-cycle implementations;		5	0	0
Control Unit design: hardwired and micro-programmed; Hazards; Exceptions; Reduced Instruction Set Computers; RISC Processor;		3		
Part – B (30 Marks)				
Pipeline: pipelined data path and control, superscalar and dynamic pipelining;	CLO – 1, 2, & 3	2	0	0
Parallel Processing: Instruction-Level Parallelism and Machine Parallelism, Instruction Issue Policy, Register Renaming, Machine Parallelism, Branch Prediction;		5	0	0
Superscalar Processors: Superscalar Execution, Superscalar Implementation;		5	0	0
Parallel Organization: Multiple Processor Organizations, Symmetric Multiprocessors, Cache Coherence and the MESI Protocol, Multithreading and Chip Multiprocessors, Clusters, Non-uniform Memory Access, Vector Computation.		5	0	0
Total		45	0	0

Textbooks:

1. William Stalling, "Computer Organization and Architecture", 9th Edition, 2014
2. David A Patterson, "Computer Organization and Design", 4th Edition, 2009
3. Andrew S. Tanenbaum, "Structured Computer Organization", 6th Edition, 2012

Course Assessment Pattern (Theory courses):

Bloom's Category		**Evaluations out of 100 marks**			
		CIE (50 marks)			**SEE (50marks)**
Cognitive Learning	Affective Learning	Mid-term (30)	Assignment/ Class Test (10)	Attendance Marks (10)	Written Exam (50)
Remember	-	5	-	-	5
Understand	-	-	5	-	5
Apply	-	5	-	-	5
Analyze	**-**	**5**	**-**	**-**	**15**
Evaluation	**-**	**10**	**5**	**-**	**10**
Create	**-**	**5**	**-**	**-**	**10**
x	Responding	x	x	10	
Remarks	Course teachers may change the magnitude of marks in Bloom's category (Both for CIE and SEE), but he/she will have to keep in mind that the % of **Higher Order Thinking Skills (HOTS)** must be about 60% or more and all the Bloom's categories to be addressed during the semester. If necessary, a course teacher may also use Cognitive (Knowledge), Affective (Attitude) and Psychomotor (Skills) domain of Bloom's Taxonomy. Question papers for semester end examination (SEE) shall be moderated by the **examination committee composed of both *internal faculty and external member***.				

Note: CIE=Continuous Internal Evaluation, SEE= Semester End Examination

i. Delivery methods & activities: Lecture, White Board Writing, Questions and Answers, **Tutorial,** Discussions Power point Presentation,

ii. **Assessment tools**: Class Attendance, Class test, Quizzes/ Assignment on problem solution, Mid-Term & Semester End Examination. Project evaluation & Viva.